

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (11/12)

1. Consider the following MIPS assembly program running on the **R10K-like processor** discussed in class, with a 7-entry reorder buffer and 13 physical registers:

```
lw    $t0, 0($s2)
and   $s2, $t2, $t1
or    $s1, $s1, $t2
sub   $t2, $s0, $s2
addi  $t0, $t0, 4
and   $s2, $s1, $s0
sub   $t0, $t1, $s1
or    $s1, $t2, $t0
sw    $s1, 4($s1)
```

Assume that **initially**: the reservation stations and reorder buffer are empty, and the contents of the map table and free list are as follows. Registers are popped from the left of the free list and pushed back onto the right.

Map Table		
Architectural Reg	Physical Reg	Ready (Y/N)
\$t0	P1	Y
\$t1	P2	Y
\$t2	P3	Y
\$s0	P4	Y
\$s1	P5	Y
\$s2	P6	Y

Free List	P7, P8, P9, P10, P11, P12, P13
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Assume that memory operations (i.e., loads and stores) take **5 cycles** to execute (i.e., 5-cycle X stage), while all other instructions take **1 cycle** to execute. The pipeline diagram of the first 12 cycles is provided below.

insn\cycle	1	2	3	4	5	6	7	8	9	10	11	12
lw \$t0, 0(\$s2)	F	De	Di	S	X	X	X	X	X	C	R	
and \$s2, \$t2, \$t1		F	De	Di	S	X	C	R	R	R	R	R
or \$s1, \$s1, \$t2			F	De	Di	S	X	C	R	R	R	R
sub \$t2, \$s0, \$s2				F	De	Di	S	X	C	R	R	R
addi \$t0, \$t0, 4					F	De	Di	S	S	S	X	C
and \$s2, \$s1, \$s0						F	De	Di	S	X	C	R
sub \$t0, \$t1, \$s1							F	De	Di	S	X	C
or \$s1, \$t2, \$t0								F	De	Di	Di	S
sw \$s1, 4(\$s1)									F	De	De	Di

- a) Fill in the contents below of the reservation stations, map table, free list and reorder buffer **at the end of cycle 8** (highlighted in red in the pipeline diagram above). Some contents are provided for you; **you only need to fill in the yellow-highlighted cells**.

Reservation Stations					
Functional Unit	Instruction	Operand (rs)		Operand (rt)	
		Physical Reg	Ready (Y/N)	Physical Reg	Ready (Y/N)
add(i)	addi \$t0, \$t0, 4	P7	N	-	-
sub					
and	and \$s2, \$s1, \$s0	P9	Y	P4	Y
or					
lw					
sw					

Map Table		
Architectural Reg	Physical Reg	Ready (Y/N)
\$t0	P11	N
\$t1	P2	Y
\$t2	P10	N
\$s0	P4	Y
\$s1	P9	Y
\$s2	P12	N

Free List	P13
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Reorder Buffer			
Instruction		Dest Physical Reg	Old Physical Reg
Head (Oldest)	lw \$t0, 0(\$s2)	P7	P1
	and \$s2, \$t2, \$t1	P8	P6
	or \$s1, \$s1, \$t2	P9	P5
	sub \$t2, \$s0, \$s2	P10	P3
	addi \$t0, \$t0, 4	P11	P7
	and \$s2, \$s1, \$s0	P12	P8
Tail (Youngest)			

- b) Continuing the program, fill in the contents below of the reservation stations, map table, free list and reorder buffer **at the end of cycle 12** (highlighted in green in the pipeline diagram on the first page).

Reservation Stations					
Functional Unit	Instruction	Operand (rs)		Operand (rt)	
		Physical Reg	Ready (Y/N)	Physical Reg	Ready (Y/N)
add(i)					
sub					
and					
or					
lw					
sw	sw \$s1, 4(\$s1)	P1	N	P1	N

Map Table		
Architectural Reg	Physical Reg	Ready (Y/N)
\$t0	P13	Y
\$t1	P2	Y
\$t2	P10	Y
\$s0	P4	Y
\$s1	P1	N
\$s2	P12	Y

Free List	P6
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Reorder Buffer			
Instruction		Dest Physical Reg	Old Physical Reg
Head (Oldest)	or \$s1, \$s1, \$t2	P9	P5
	sub \$t2, \$s0, \$s2	P10	P3
	addi \$t0, \$t0, 4	P11	P7
	and \$s2, \$s1, \$s0	P12	P8
	sub \$t0, \$t1, \$s1	P13	P11
	or \$s1, \$t2, \$t0	P1	P9
Tail (Youngest)	sw \$s1, 4(\$s1)	-	-

- c) Why does the **addi \$t0, \$t0, 4** instruction **wait** at the Issue (S) stage in cycle 8? Select all the reasons that apply.
- i. The reorder buffer is full.
 - ii. Register operands are not all ready
 - iii. There is no available reservation station.
 - iv. The register free list is empty.

Answer:

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- d) Why does the **or \$s1, \$t2, \$t0** instruction **stall** at the Dispatch (Di) stage in cycle 10? Select all the reasons that apply.
- i. The reorder buffer is full.
 - ii. Register operands are not all ready
 - iii. There is no available reservation station.
 - iv. The register free list is empty.

Answer:

i and iv